

CLOUD COMPUTING- CSA1506

Assessment Tool-2-C03

QUESTION: 1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Answer:

Introduction:

During seasonal sales such as Black Friday, Diwali, or festival offers, e-commerce websites often receive millions of visitors within a short period. If the infrastructure cannot handle the increased demand, the website may become slow or unavailable. Cloud computing solves this problem using Auto-Scaling, Load Balancing, and Monitoring Tools.

1. Cloud Auto-Scaling

Auto-scaling is a cloud feature that automatically increases or decreases the number of servers (virtual machines or containers) based on current workload.

Working:

- During high traffic:
 - New server instances are automatically created.
 - Incoming users are served without delays.
- During low traffic:
 - Extra servers are removed.
 - This reduces infrastructure cost.

Benefits:

- Handles sudden traffic spikes.
- Prevents server overload.
- Reduces operational costs.

- Maintains application performance.
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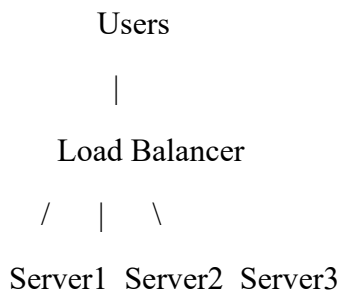
2. Load Balancing

A **Load Balancer** distributes incoming user requests evenly across multiple servers.

Working:

- Receives all user requests.
- Checks which server has less workload.
- Sends requests to the best available server.

Example:



Advantages:

- Prevents one server from becoming overloaded.
 - Improves response time.
 - Ensures high availability.
 - Automatically redirects traffic if one server fails.
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3. How Auto-Scaling and Load Balancing Work Together

When traffic suddenly increases:

1. Monitoring tools detect high CPU usage.
2. Auto-scaling launches additional servers.
3. Load balancer immediately starts sending requests to the new servers.
4. Users experience fast and uninterrupted service.

When traffic decreases:

- Auto-scaling removes unnecessary servers.
- Load balancer distributes requests among the remaining servers.

This ensures efficient resource utilization.

4. Reduction of Downtime

Without cloud services:

- Server crashes due to overload.
- Website becomes unavailable.
- Customers cannot place orders.

With auto-scaling and load balancing:

- Extra servers are added automatically.
- Failed servers are bypassed.
- Service continues without interruption.

Hence, downtime is greatly reduced.

5. Improvement of User Experience

Users benefit from:

- Faster page loading.
- Smooth browsing.
- Quick checkout process.
- Reduced waiting time.
- Better reliability.

A good user experience increases customer satisfaction and sales.

6. Role of Monitoring Tools

Monitoring tools continuously observe the health and performance of cloud resources.

They monitor:

- CPU utilization
- Memory usage
- Disk usage
- Network traffic

- Response time
- Error rates
- Number of active users

Functions:

- Detect performance issues.
- Trigger auto-scaling policies.
- Send alerts to administrators.
- Generate reports and logs.
- Help identify failures quickly.

Examples:

- Amazon CloudWatch
 - Google Cloud Monitoring
 - Azure Monitor
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7. Real-World E-commerce Scenario

Consider an online shopping company such as **Amazon** during the **Great Indian Festival Sale**.

Situation:

- Millions of customers visit the website simultaneously.
- Product searches, payments, and orders increase rapidly.

Cloud Solution:

- Auto-scaling automatically launches additional servers as traffic increases.
- Load balancer distributes customer requests evenly across all servers.
- Monitoring tools continuously monitor CPU usage, response time, and network traffic.
- If any server becomes unhealthy, traffic is redirected to healthy servers.

Result:

- Website remains available.
- Pages load quickly.
- Customers complete purchases without interruption.

- The company avoids revenue loss and maintains customer satisfaction.
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Advantages

- Automatically handles traffic spikes.
 - High availability.
 - Reduced downtime.
 - Better application performance.
 - Cost-efficient resource usage.
 - Improved customer experience.
 - Reliable and scalable infrastructure.
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Conclusion

Cloud auto-scaling, load balancing, and monitoring tools work together to ensure that applications remain available and responsive during sudden traffic spikes. Auto-scaling adjusts computing resources based on demand, load balancing distributes requests efficiently, and monitoring tools continuously track system health and trigger scaling actions. This combination minimizes downtime, improves user experience, and enables e-commerce companies to handle large seasonal sales successfully.

